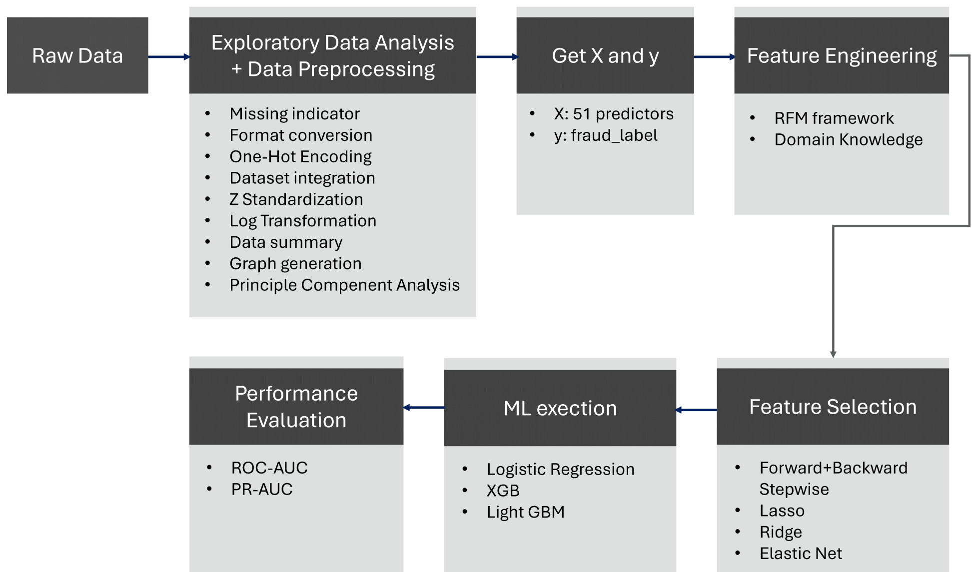
Graphical Abstract



Highlights

Integrating Enhanced RFM-Based Feature Engineering and Selection for Credit Card Fraud Detection

WEI-PENG CHEN, LIN YICHEN, HUI HSIN WANG, HUEI-WEN TENG

- Integrated behavioral feature engineering using the Recency–Frequency–Monetary (RFM) framework together with domain knowledge and exploratory data analysis (EDA), producing structured indicators that capture meaningful fraud-related activity patterns.
- Evaluated model performance using ROC-AUC and PR-AUC across Logistic Regression, XGBoost, and LightGBM, showing that engineered behavioral features substantially improve predictive accuracy compared to traditional feature selection alone.
- Findings demonstrate that RFM and domain-knowledge features enhance both interpretability and generalization, reducing overfitting in gradient boosting models while providing actionable behavioral signals for fraud detection.

Integrating Enhanced RFM-Based Feature Engineering and Selection for Credit Card Fraud Detection

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Abstract

Detecting fraudulent credit card transactions remains challenging due to the behavioral complexity of fraud and its rarity within large volumes of legitimate activity. This study adopts a feature-centered approach, combining an extended Recency–Frequency–Monetary (RFM) framework with domain knowledge and exploratory data analysis to construct informative behavioral features. Feature selection is implemented through wrapper-based subset selection technique and embedding-based shrinkage approaches to refine the predictor set. Empirical evaluation using Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and Precision–Recall Area Under the Curve (PR-AUC) demonstrates that the combined feature engineering and selection approach improves the model's capacity to identify fraud-related signals under highly imbalanced conditions. These findings underscore the value of feature-focused modeling strategies for enhancing fraud detection performance in real-world payment environments.

Keywords: Fraud Detection, Feature Engineering, Feature Selection
JEL: C45, C53, G21, K42, M41

1. Introduction

1.1. Background and Importance of Credit Card Fraud

Credit card fraud has become a significant global concern, generating substantial financial losses and reducing consumer confidence in payment systems (Ahmed, 2023). According to the Federal Trade Commission (FTC), reported identity and payment fraud cases in the United States reached 458,538 in 2024 (Federal Trade Commission, 2025). In Europe, combined fraud losses documented by the European Banking Authority (EBA) and the European Central Bank (ECB) exceeded €4.3 billion in 2022, with an additional €2 billion in early 2023 (European Banking Authority, 2024). Taiwan has experienced a similar rise, with fraudulent credit card transactions totaling NT$3.2 billion in 2023 and projected to continue increasing in 2024 (National Credit Card Center, 2025). These trends highlight the need for analytical frameworks capable of detecting evolving fraud behaviors within highly skewed datasets.

1.2. Research Directions in Fraud Detection

Prior research in credit card fraud detection has explored multiple methodological directions. A substantial portion of the literature focuses on handling imbalanced datasets through oversampling and undersampling techniques, expanding the representation of minority fraudulent records (Chawla et al., 2002; He et al., 2008). Other studies introduce algorithm-level strategies such as ensemble models and cost-sensitive learning to address skewness more directly (Haixiang et al., 2017). These approaches form an important part of the broader landscape of fraud detection research.

Complementary to data-level strategies, improving feature representation is critical for capturing nuanced fraud behaviors. The Recency–Frequency–Monetary (RFM) framework provides an interpretable behavioral structure for constructing fraud-related variables (Baesens et al., 2021). Its extensions, including APATE (Van Vlasselaer et al., 2015) and HOBA (Zhang et al., 2021), incorporate additional contextual indicators such as geographical, temporal, and network-based information, enriching the feature space used in detection models.

Feature selection is commonly categorized into three classes: Filter, Wrapper, and Embedding(Tang et al., 2014). To refine the predictor space in this study, Stepwise selection is applied as a Wrapper-based subset selection technique, while Lasso, Ridge, and Elastic Net regularization are used as

Embedding-based shrinkage approaches. These methods, combined with domain-informed and EDA-derived features, aim to improve the model's sensitivity to fraud signals within highly imbalanced transaction data.

1.3. Purpose of This Study

Building on a feature-focused perspective, this paper proposes a framework that integrates enriched RFM-based feature engineering with systematic feature selection. The combination of behavioral indicators, domain knowledge, and exploratory insights allows for a more informative predictor set, which is then refined through Wrapper and Embedding selection methods to enhance fraud detection performance.

Model performance is evaluated using Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and Precision–Recall Area Under the Curve (PR-AUC), metrics particularly suitable for rare-event classification.

1.4. Organization of This Paper

The remainder of this paper is organized as follows. Section 2 describes the dataset and presents exploratory data analysis. Section 3 introduces logistic regression as the baseline model and details the proposed feature engineering and selection framework. Section 4 reports empirical results based on ROC-AUC and PR-AUC. Section 5 concludes with key findings and potential areas for future work.

2. Data

2.1. Data exploration

The dataset used in this study, summarized in Table 1, was obtained from Kaggle and includes credit card transactions recorded between January 1, 2010, and October 31, 2019, totaling 13,305,915 entries, of which 8,914,963 contain fraud labels. Originally distributed across five source tables—`Transaction`, `User`, `Credit Card`, `Train_Fraud_Labels`, and `MCC_Codes`—the data consisted of 37 columns. After integration and pre-processing, including one-hot encoding for categorical variables and missing value imputation, the dataset expanded to 51 columns. Table 2 summarizes the variables prior to integration. The target variable, `fraud_label`, indicates whether a transaction is fraudulent. The dataset exhibits an overall fraud rate of 0.100% and 0.149% within the labeled subset, reflecting a severe class imbalance that complicates model training and evaluation.

Table 1: Description of dataset

Dataset	Samples (Labeled Samples)	columns	Fraud	Default rate
Transactions	13,305,915 (8,914,963)	37	13,332	0.10%(0.14%)

The dataset contains 51 columns of various types, including 7 categorical variables, 3 date-time variables, 28 numerical variables (26 integers and 2 floating-point values), 3 object/text variables, and 10 binary variables.

#	Column	**Dtype**	**Description**
0	`transaction_id`	int64	Unique ID for each transaction
1	`date`	datetime64[ns]	Timestamp of the transaction.
2	`client_id`	int64	Unique ID for each client
3	`card_id`	int64	Unique ID for each card
4	`amount`	int64	Transaction amount
5	`use_chip`	category	Transaction method
6	`merchant_id`	int64	Unique merchant identifier
7	`merchant_city`	category	City where merchant is located
8	`merchant_state`	category	State or region of the merchant
9	`zip`	int64	ZIP or postal code of merchant
10	`mcc_code`	int64	Merchant Category Code (MCC)
11	`errors`	category	Transaction error or anomaly type
12	`merchant_state_missing_flag`	int64	Flag for missing merchant state
13	`zip_missing_flag`	int64	Flag for missing ZIP code
14	`errors_missing_flag`	int64	Flag for missing error data
15	`use_chip_Chip Transaction`	uint8	Indicator for chip-based transactions
16	`use_chip_Online Transaction`	uint8	Indicator for online transactions
17	`use_chip_Swipe Transaction`	uint8	Indicator for swipe transactions
18	`fraud_label`	bool	Whether the transaction is fraudulent

#	Column	**Dtype**	**Description**
19	`current_age`	int64	Current age of the client
20	`retirement_age`	int64	Expected retirement age
21	`birth_year`	int64	Client's birth year
22	`birth_month`	int64	Client's birth month
23	`gender`	category	Gender of the client
24	`address`	object	Client's registered address
25	`latitude`	float64	Latitude of client address
26	`longitude`	float64	Longitude of client address
27	`per_capita_income`	int64	Client's annual capital income
28	`yearly_income`	int64	Client's annual income
29	`total_debt`	int64	Total outstanding debt amount
30	`credit_score`	int64	Client's creditworthiness score
31	`num_credit_cards`	int64	Number of credit cards owned by client
32	`card_brand`	category	Brand of the card
33	`card_type`	category	Type of card
34	`card_number`	int64	Primary account number
35	`expires`	datetime64[ns]	Card expiration date
36	`cvv`	int64	Masked card security code
37	`has_chip`	int64	Whether the card includes a chip
38	`num_cards_issued`	int64	The number of specific credit cards issued under a client's account
39	`credit_limit`	int64	Credit limit assigned to the card
40	`acct_open_date`	datetime64[ns]	Date when the account was opened
41	`year_pin_last_changed`	int64	Last year the PIN was updated
42	`card_on_dark_web`	object	Whether card info appears on dark web
43	`card_type_Credit`	uint8	Indicator for credit card type
44	`card_type_Debit`	uint8	Indicator for debit card type
45	`card_type_Debit (Prepaid)`	uint8	Indicator for prepaid debit cards
46	`card_brand_Amex`	uint8	Indicator for Amex cards

#	Column	Dtype	Description
47	`card_brand_Discover`	uint8	Indicator for Discover cards
48	`card_brand_Mastercard`	uint8	Indicator for Mastercard cards
49	`card_brand_Visa`	uint8	Indicator for Visa cards
50	`fraud_label_missing_flag`	int64	Flag for missing fraud label

Table 2: Dataset description for credit card transactions and client information.

2.2. Data cleaning, and data pre-processing

During the pre-processing stage, several procedures were applied, including missing indicator creation, format conversion, one-hot encoding, dataset integration, Z-standardization, and log transformation.

Among all columns, the amount variable exhibits the most severe skewness and kurtosis, with values of 5.226846 and 111.279744, respectively. To address this issue, we applied several transformation techniques, including log transformation, square root, cube root, and Yeo–Johnson transformations. After comparison, the cube root transformation demonstrated the best performance and was therefore adopted in subsequent analyses.

Table 3: Transformation comparison of skewness and kurtosis

Transformation	**Skewness**	**Kurtosis**
Original	5.226846	111.279744
Log(x+1, negatives→0)	-0.559664	-0.358987
Square root (negatives→0)	1.283660	5.022708
Cube root	-1.705728	5.084572
Box–Cox (skip: nonpositive data)	NaN	NaN
Yeo–Johnson ($\lambda = 0.976$)	3.658878	83.135841

Moreover, The dataset contained 4,390,952 missing values in the `fraud_label` column, accounting for approximately one-third (33.0%) of the original data. Upon examination, these missing values were determined to be missing completely at random (MCAR). Consequently, subsequent EDA was conducted using only the records with available `fraud_label` values.

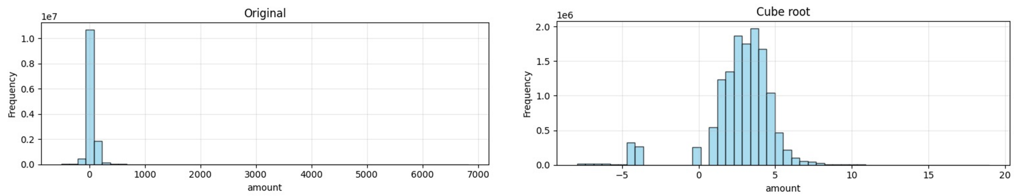

Figure 1: Amount Transformation

2.3. Cross-sectional data

The exploratory analysis began with a cross-sectional examination of the features to understand their individual and interactive relationships with the target variable, `fraud_label`.

2.3.1. Analysis of numerical variables

Box plots (Figure 2) and scatter plots/matrix (Figure 3) for the numerical variables are presented. The scatter matrix reveals a critical interaction: the fraudulent transactions (blue dots) predominantly occur in a high-risk region defined by a combination of high log-transformed transaction amount (`log(amount)`) and a low credit limit (`credit_limit`).

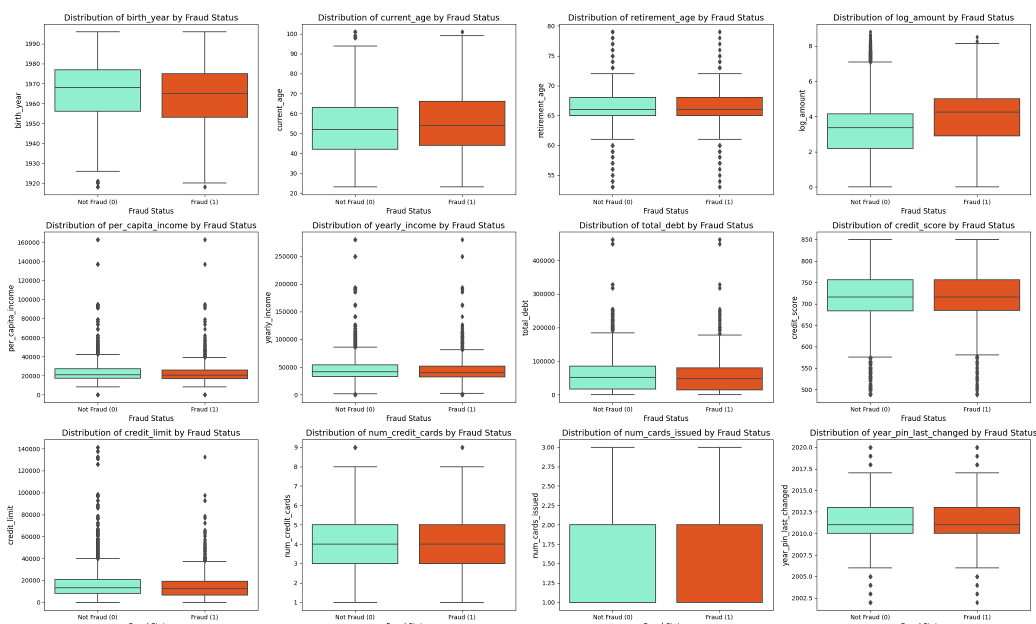

Figure 2: Box plot for numerical variables

The scatter matrix plot indicates strong correlations between `birth_year` and `current_age`, as well as between `per_capita_income` and `yearly_income`. Preliminary analysis began with an examination of inter-feature correlations. The scatter matrix plot revealed strong, unsurprising linear relationships, specifically between `birth_year` and `current_age`, and between `per_capita_income` and `yearly_income`.

2.3.2. Initial Feature-to-Target Correlation

A visual inspection of the fraud rate (calculated as the ratio of `fraud_label` to transaction counts) across different levels of numerical variables, as presented in Figure 4, suggested several key relationships with the target variable. Specifically, variables such as `log_amount`/`amount`, `num_credit_card`, and `num_cards_issued` appear to exhibit a positive correlation with the fraud rate. Conversely, the `credit_limit` variable demonstrated a clear negative correlation, confirming its expected protective effect against fraudulent transactions.

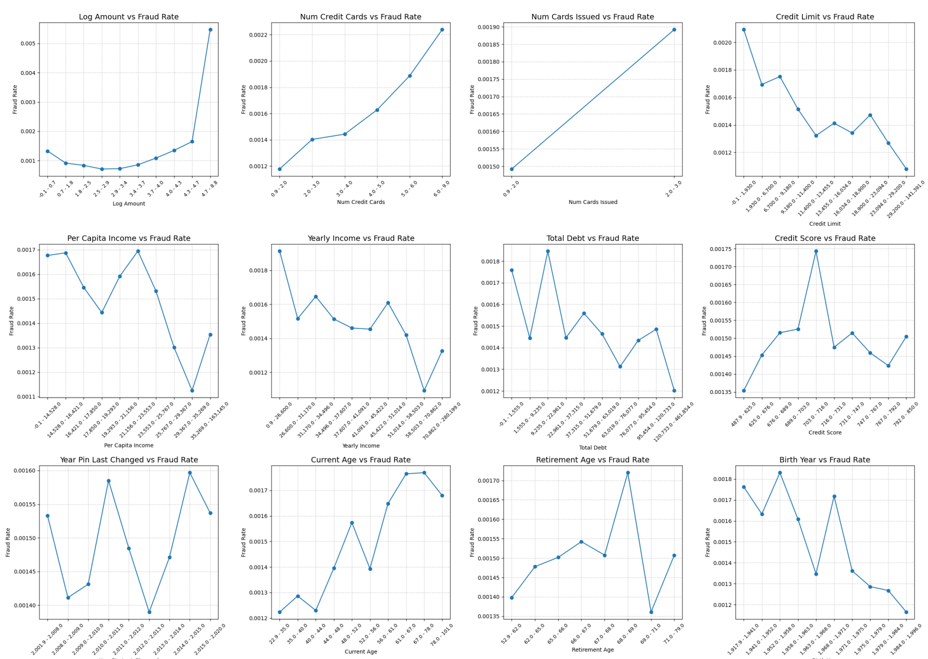

Figure 4: Fraud rate across different levels of numerical variables.

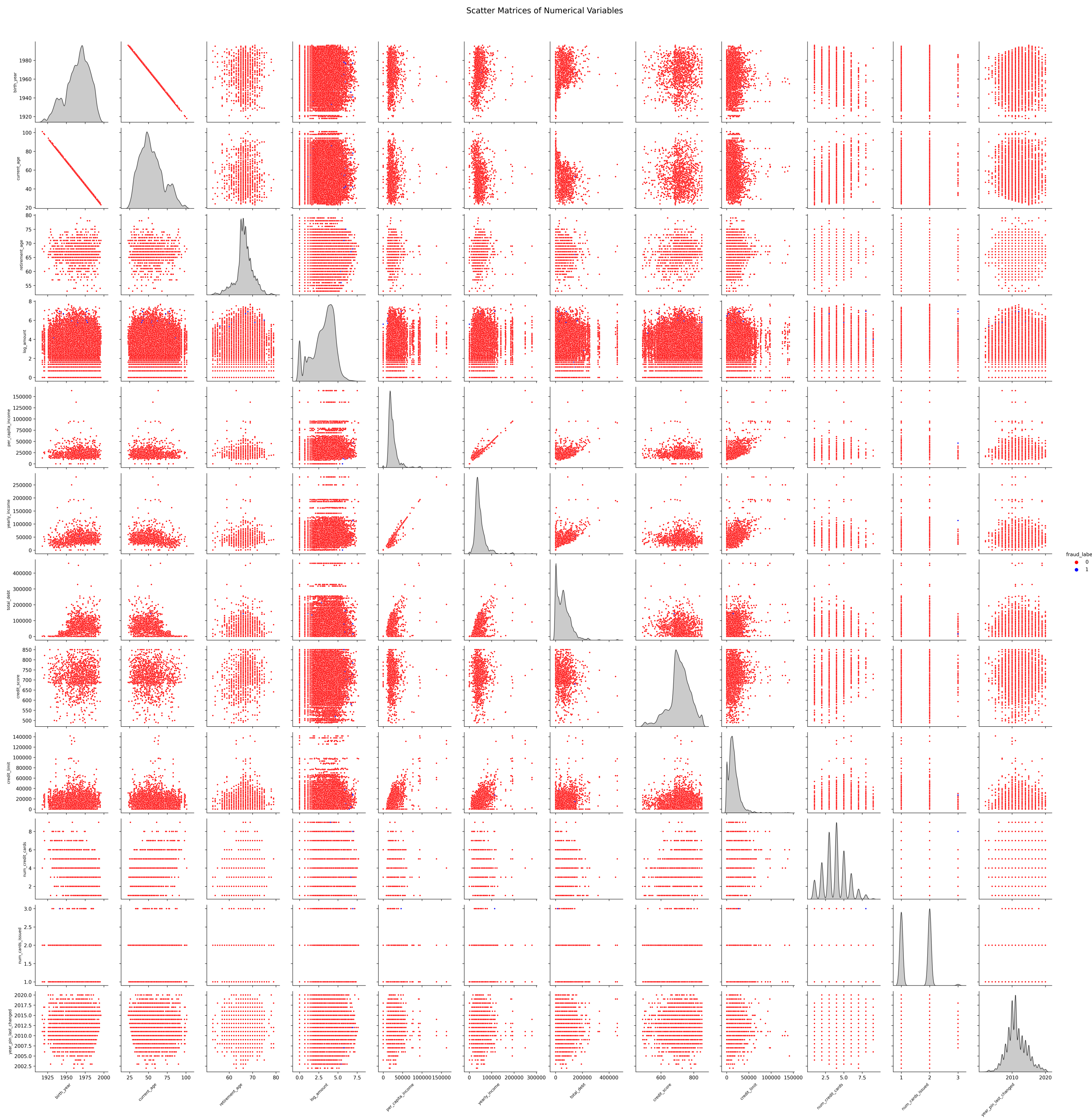

Figure 3: Scatter matrix for numerical variables

2.3.3. Feature Correlation and Engineering Hypotheses

Further analysis using heatmaps was conducted to formally assess collinearity (Figure 5 and Figure 6).

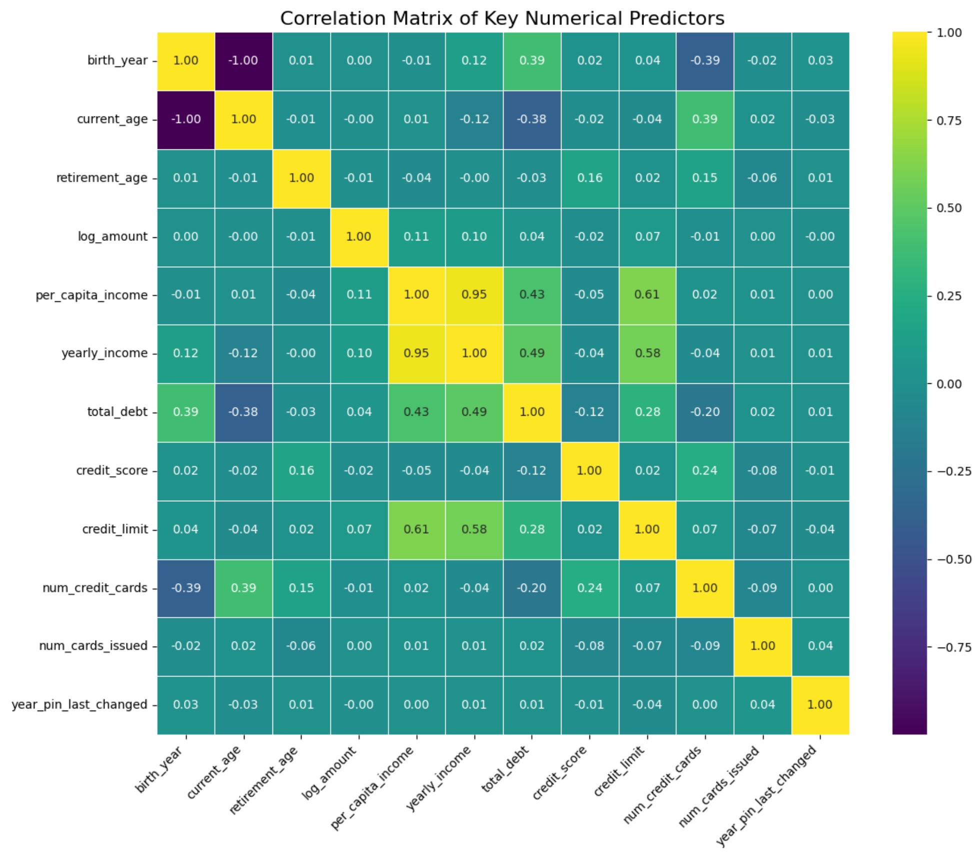

Figure 5: Numerical variable correlation heatmap

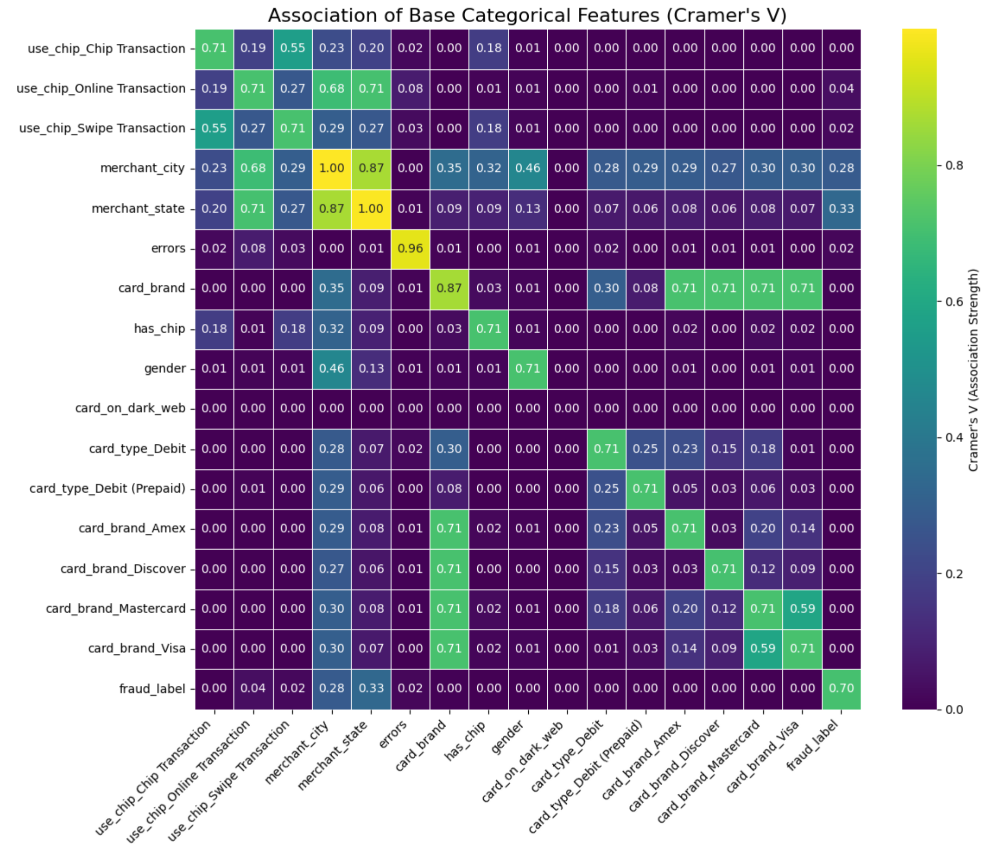

Figure 6: Categorical variable correlation heatmap

The numerical correlation heatmap (Figure 5) indicated that, apart from the expected definitional relationships (e.g., `current_age` vs. `birth_year`), there was low collinearity among the independent features. This minimal redundancy is favorable for maintaining model stability. For categorical variables, a heatmap (Figure 6) was generated using Cramér's V to measure association. This analysis also did not indicate any prominent correlation coefficients; however, even modest associations ($V < 0.2$) in a fraud context can be highly predictive.

The distinct patterns observed in the correlation analysis, particularly the inverse relationship involving `credit_limit`, strongly motivated the development of an interaction feature. The most powerful feature hypothesis generated by this analysis is the Transaction-to-Limit Ratio (`amount/credit_limit`). This ratio is anticipated to be highly discriminatory because fraudulent transactions are typically characterized by an attempt to utilize a high proportion of the available credit limit in a single event.

These findings establish a solid empirical foundation for subsequent advanced feature engineering, which will include Target Encoding highly-cardinality categorical variables (such as `merchant_id` or `merchant_city`) to transform their inherent, category-specific fraud rates into robust predictive numerical features for the final model construction.

3. Methodology

After defining the dependent variable X_{18} `fraud_label`, logistic regression is adopted as the benchmark model due to its interpretability and relatively mild modeling assumptions.

All numerical and categorical independent variables are considered for model development. To ensure that the assumptions of logistic regression are satisfied, several diagnostic checks are conducted. Because each transaction in the public dataset is recorded independently, we assume independence among observations. In addition, multicollinearity is assessed using the variance inflation factor (VIF), and variables are iteratively examined until all VIF values fall below 4, thereby reducing the likelihood of multicollinearity.

Building on the benchmark model, we further perform feature engineering and propose an enhanced RFM-based framework to generate additional features. Multiple model specifications—including traditional logistic regression, XGBoost (XGB), and LightGBM (LGBM)—are evaluated to investigate whether these engineered features can improve classification performance relative to the baseline logistic regression classifier. Details of the research design and methodology are presented in Section 3.2.

Finally, model evaluation incorporates both statistical and predictive perspectives. Given the substantial class imbalance in the dataset, we prioritize AUC and PR-AUC over conventional accuracy metrics when comparing model performance. The overall modeling workflow is illustrated in Figure 7.

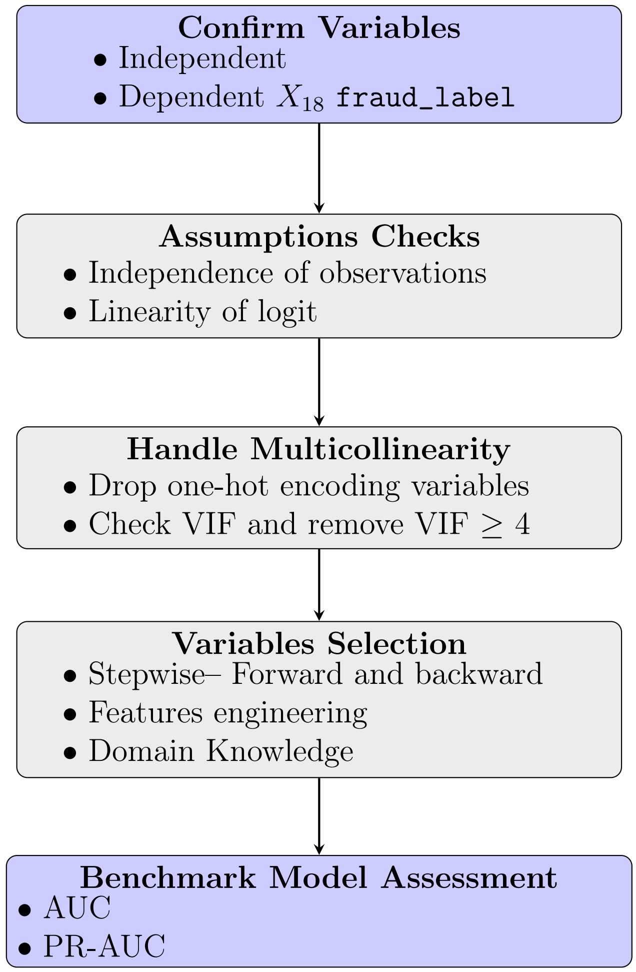

Figure 7: Workflow of Logistic Regression Model Building

3.1. Benchmark Model—Logistic Regression

The logistic regression model estimates the probability that the dependent variable Y equals 1 given a set of predictors $X_1, X_2, \ldots, X_k$. After applying the logit transformation, the model is expressed as:

$$\text{logit}[\pi(X)] = \ln\left(\frac{\pi(X)}{1-\pi(X)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k,$$

where $\pi(X) = P(Y = 1 \mid X_1, X_2, \ldots, X_k)$.

Logistic regression is selected as the benchmark model because the target variable is binary, and the method provides clear interpretability with fewer distributional assumptions relative to other machine learning classifiers.

The main assumptions include independence of observations, linearity between the logit of the outcome and continuous predictors, and the absence of severe multicollinearity. Unlike discriminant analysis, logistic regression does not require normality of predictors.

A sufficient sample size is also necessary to ensure stable estimation. A commonly used guideline requires

$$n \geq 400 \quad \text{and} \quad n \geq 100 \times (\text{number of independent variables}).$$

The following subsections describe the model construction procedures, including assumption checks and the assessment of data adequacy.

3.1.1. Stage 1: Confirm Variables

Among the 51 available variables in the dataset, both categorical and numerical features were considered as potential independent variable candidates for model development. In total, 37 variables (20 numerical and 17 categorical) were included for further analysis, as summarized in Table 4.

Table 4: Summary of Dependent and Candidate Independent Variables

Variables	dtype	Variables	dtype
X_{19} fraud_label **(Dependent)**	num.	X_{28} per_capita_income	num.
X_1 transaction_id	num.	X_{29} yearly_income	num.
X_3 client_id	num.	X_{30} total_debt	num.
X_4 card_id	num.	X_{31} credit_score	num.
X_5 amount	num.	X_{32} num_credit_cards	num.
X_7 merchant_id	num.	X_{35} card_number	num.
X_{10} zip	num.	X_{37} cvv	num.
X_{11} mcc_code	num.	X_{38} has_chip	cate.
X_{13} merchant_state_missing_flag	cate.	X_{39} num_cards_issued	num.
X_{14} zip_missing_flag	cate.	X_{40} credit_limit	num.
X_{15} errors_missing_flag	cate.	X_{44} card_type_Credit	cate.
X_{16} use_chip_Chip_Transaction	cate.	X_{45} card_type_Debit	cate.
X_{17} use_chip_Online_Transaction	cate.	X_{46} card_type_Debit (Prepaid)	cate.
X_{18} use_chip_Swipe_Transaction	cate.	X_{47} card_brand_Amex	cate.
X_{20} current_age	num.	X_{48} card_brand_Discover	cate.
X_{21} retirement_age	num.	X_{49} card_brand_Mastercard	cate.
X_{24} gender	cate.	X_{50} card_brand_Visa	cate.
X_{26} latitude	num.	X_{51} fraud_label_missing_flag	cate.
X_{27} longitude	num.		

3.1.2. Stage 2: Assumptions Checks

The independence of observations cannot be strictly verified because multiple transactions may originate from the same customer. In the absence of hierarchical identifiers or clustering structures, each transaction is treated as an independent observation. This assumption is therefore pragmatically adopted for model estimation.

Linearity in the logit was assessed using the Box–Tidwell test. Several variables were found to violate this assumption, including X_1 `transaction_id`, X_5 `amount`, X_7 `merchant_id`, X_{10} `zip`, X_{11} `mcc_code`, X_{20} `current_age`, X_{21} `retirement_age`, and X_{35} `card_number`. Multiple transformation attempts did not resolve the nonlinearity. Given their practical relevance and demonstrated predictive contribution, these variables were retained in the model.

Sample size adequacy was also examined to ensure stable parameter estimation. Both the training and test sets satisfied the commonly applied guideline:

$$n \geq 400 \quad \text{and} \quad n \geq 100 \times 37 \text{ (number of independent variables).}$$

The training set contained 7,131,970 observations and the test set contained 1,782,994 observations, both far exceeding the minimum threshold.

3.1.3. Stage 3: Handling Multicollinearity

Violations of the multicollinearity assumption may inflate standard errors and reduce the reliability of coefficient estimates, thereby weakening both interpretability and predictive stability. Accordingly, assessing the contribution of each variable and removing predictors that introduce excessive collinearity is essential to ensure robust parameter estimation and overall model consistency.

3.1.4. Stage 4: Variable Selection

To obtain a parsimonious yet informative specification, this stage implements a bidirectional stepwise selection procedure within the logistic regression framework. The primary criterion for variable entry is improvement in overall model fit, assessed through the reduction in the -2 Log Likelihood (2LL) statistic. A candidate predictor is incorporated only if its inclusion yields the greatest and statistically significant decrease in the 2LL value, thereby contributing meaningfully to the model's explanatory power.

The stepwise algorithm alternates between forward inclusion and backward elimination. During the forward phase, variables not currently in the model are individually evaluated, and the predictor that produces the largest reduction in 2LL beyond the predefined inclusion threshold (α_{in}) is added. In the backward phase, all selected variables are reassessed, and any predictor whose removal does not significantly deteriorate the 2LL fit—or whose p-value exceeds the elimination threshold (α_{out})—is removed.

This iterative process continues until no additional improvement in 2LL can be achieved or until a pre-specified limit on the number of predictors (k) is reached. By prioritizing the reduction in the 2LL statistic rather than relying solely on significance levels, the procedure yields a final model that balances explanatory strength, parsimony, and statistical stability.

In addition to the stepwise procedure, we further integrate an enhanced RFM-based feature engineering framework and domain-informed constructed variables into the pool of candidate predictors(see Section 3.2.1 Feature

Engineering). These engineered features are incorporated into subsequent model development to assess their capacity to improve predictive performance relative to the benchmark logistic regression model.

3.1.5. Stage 5: Benchmark Model Assessment

In the final stage, the logistic regression model was evaluated primarily using threshold-independent performance metrics, given the highly imbalanced nature of the fraud detection dataset. In particular, the Area Under the Receiver Operating Characteristic Curve (AUC) and the Area Under the Precision-Recall Curve (PR-AUC) were employed as the main criteria for model assessment.

The ROC–AUC provides a comprehensive measure of the model's ability to discriminate between fraudulent and legitimate transactions across all possible classification thresholds. Complementarily, the PR-AUC emphasizes performance on the minority class and is particularly informative in imbalanced settings, reflecting the trade-off between precision and recall.

By focusing on AUC and PR-AUC, this evaluation framework offers a robust and threshold-independent appraisal of the model's discriminative power, ensuring that the model's effectiveness in detecting rare fraudulent events is accurately captured.

3.2. Design of Research

This study adopts a structured research design to investigate the impact of feature engineering on model performance. The design comprises two main components. First, feature engineering involves the creation of new variables informed by the RFM framework, domain knowledge and EDA, which are subsequently organized into distinct feature sets for model training and evaluation. Second, model performance is assessed using well-defined metrics, with the dataset divided into training and testing subsets to ensure robust evaluation and prevent overfitting.

3.2.1. Feature Engineering

Building on the RFM-inspired design (Table 5), we define new features as follows: Recency features quantify the time intervals between consecutive transactions, Frequency features count the number of transactions within 7-, 30-, 60-, and 90-day windows, and Monetary features measure changes in transaction amounts over time.

Table 5: RFM-based feature dimensions capturing Recency, Frequency, and Monetary aspects of user transactions.

Dimension	**Feature Name**	**Definition**
Recency	`RecencyInterval`	Time interval between consecutive transactions.
Frequency	`TxnFrequency_`X`d`	Number of transactions within an X-day window (e.g., 7, 30, 60, 90 days).
Monetary	`AmtDelta`	Change in average transaction amounts across time.

Building on research showing that spatial and contextual factors can enhance RFM-based features, we incorporate location-related variables (e.g., online vs. offline transactions, first transactions within a city, state, or region, and high-risk Merchant Category Codes) as well as behavior-driven indicators such as the transaction-to-limit ratio (TxnToLimitRatio), which captures spending intensity relative to the cardholder's available credit. Together, these engineered features provide a comprehensive, user-centric representation of transactional patterns, improving the model's ability to detect anomalous and potentially fraudulent behavior. Table 6 summarizes these features and their definitions.

Table 6: Feature dimensions related to location-based and other behavioral indicators.

Dimension	Feature Name	Definition
Location	merchant_[location]	Indicates the transaction's location (e.g., online, US, EU, others).
Location	FirstTxnInRegion	Indicates whether the transaction is the first occurrence in a specific city, state, or region.
Location	HighRiskMCC	Flags transactions associated with high-risk Merchant Category Codes (MCC).
Location	DifferentState	Indicates whether the transaction occurred in a different state compared to the previous transaction for the same card.
Others	TxnToLimitRatio	Ratio of the transaction amount to the cardholder's credit limit.

To assess the impact of feature engineering, we constructed three distinct feature groups for experimental evaluation: the RFM-based set X_{rfm}, the domain-knowledge-based set X_{dk}, and the comprehensive set X_{raw}. For X_{raw}, a multi-stage feature selection process was applied to remove variables according to predefined rules, including the elimination of target variables and identifiers, exclusion of post-hoc or data quality indicators, removal of categorical features, elimination of highly collinear features as indicated by the Variance Inflation Factor (VIF), and removal of date-time and other miscellaneous variables. The resulting set comprises 36 predictors. Table 7 presents the variables grouped accordingly.

Table 7: Grouping for Features

Group	Features	
X_{raw}	`amount`	`current_age`
	`retirement_age`	`yearly_income`
	`total_debt`	`credit_score`
	`num_credit_cards`	`has_chip`
	`num_cards_issued`	`credit_limit`
	`year_pin_last_changed`	
X_{rfm}	`RecencyInterval`	`TxnFrequency_7d`
	`TxnFrequency_30d`	`TxnFrequency_60d`
	`TxnFrequency_90d`	`AmtDelta`
X_{dk}	`merchant_online`	`merchant_us`
	`merchant_eu`	`merchant_others`
	`FirstTxnInRegion`	`HighRiskMCC`
	`TxnToLimitRatio`	`DifferentState`

3.2.2. Performance measures

To comprehensively and objectively evaluate the performance of various classification models and their corresponding feature sets (X_{raw}, X_{dk}, X_{rfm}), we will utilize two critical Area Under the Curve (AUC) metrics: the Receiver Operating Characteristic AUC (ROC-AUC) and the Precision-Recall AUC (PR-AUC). The ROC-AUC serves as the standard measure of a model's discriminative power, reflecting its overall ability to distinguish between positive and negative classes across varying thresholds. Since it is relatively insensitive to changes in class distribution, it is ideal for initial, high-level comparisons of model potential.

However, given the context of financial fraud detection, which involves extreme class imbalance, the PR-AUC is considered the more practically relevant metric. The PR-AUC focuses specifically on the positive class (fraudulent transactions) by plotting the relationship between Precision and Recall. Therefore, the value of PR-AUC more truly reflects the model's effectiveness and robustness in identifying the rare but critical minority samples in a real-world setting. We will calculate both metrics to compare the models' fitting ability on the Training Set with their generalization ability on the Test Set, with PR-AUC serving as the primary basis for final model selection.

For model evaluation, the dataset is split chronologically, with the first

80% used for training and the remaining 20% reserved for testing. This temporal split ensures that the model is trained on past data and evaluated on future observations, thereby preventing information leakage and preserving the sequential integrity of transactions.

4. Results

We evaluated Logistic Regression, XGBoost, and LightGBM across different feature configurations to assess the impact of feature engineering and selection on predictive performance. Model performance was measured using AUC and PR-AUC on both training and testing sets.

Table 8: Model performance comparison

Index	Model	Features	AUC		PR AUC	
			Train	Test	Train	Test
0	LR	X_{all}	0.8859	0.8862	0.0193	0.0257
1	LR	forward+backward stepwise	0.8840	0.8374	0.0224	0.0256
2	LR	lasso	0.8841	0.8378	0.0223	0.0256
3	LR	ridge	0.8841	0.8378	0.0223	0.0256
4	LR	elastic net	0.8840	0.8378	0.0223	0.0256
5	LR	$X_{\text{raw}} + X_{\text{rfm}} + X_{\text{dk}}$ +elatic net	0.9408	0.9520	0.1068	0.0506
6	LR	$X_{\text{raw}} + X_{\text{rfm}} + X_{\text{dk}}$ $-X_{\text{identifiers}}$	0.9143	0.9025	0.0811	0.0695
7	XGB	X_{all}	0.9900	0.9800	0.8200	0.4200
8	XGB	$X_{\text{raw}} + X_{\text{rfm}} + X_{\text{dk}}$	0.9800	0.9900	0.3900	0.3400
9	LGBM	X_{all}	0.9900	0.9600	0.8000	0.3900
10	LGBM	$X_{\text{raw}} + X_{\text{rfm}} + X_{\text{dk}}$	0.9300	0.9600	0.3200	0.2900

Abbreviations: LR: Logistic Regression; XGB: XGBoost; LGBM: LightGBM.

4.1. Feature Selection

Logistic Regression trained on all features (X_{all}, Baseline model in Table 8) showed solid AUC values (0.8859 training, 0.8862 testing) but relatively low PR-AUC (0.0193 training, 0.0257 testing). Applying traditional feature selection methods—including forward/backward stepwise (Model 1), lasso (Model 2), ridge (Model 3), and elastic net (Model 4)—produced only minor changes in performance, indicating that feature selection alone had limited impact on Logistic Regression.

4.2. Feature Engineering

Introducing engineered features combining X_{raw}, X_{rfm}, and domain knowledge features X_{dk}, along with elastic net regularization (Model 5), led to substantial improvements. Training and testing AUC increased to 0.9408 and 0.9520, respectively, while PR-AUC also improved (0.1068 training, 0.0506 testing). After carefully removing potential identifiers to prevent data leakage (Model 6), training AUC decreased slightly to 0.9143, but testing PR-AUC improved to 0.0695, suggesting reduced overfitting. These results highlight that thoughtful feature engineering can significantly boost model performance beyond what is achieved with feature selection alone.

4.3. Model Selection

Gradient boosting models consistently outperformed Logistic Regression. XGBoost trained on all features (Model 7) achieved a training AUC of 0.9900, a testing AUC of 0.9800, and a testing PR-AUC of 0.4200. Using the engineered feature set ($X_{\text{raw}} + X_{\text{rfm}} + X_{\text{dk}}$, Model 8) slightly reduced PR-AUC to 0.3400, although testing AUC increased to 0.9900. LightGBM trained on all features (Model 9) reached comparable performance (0.9900 training, 0.9600 testing AUC; 0.3900 testing PR-AUC), while the engineered feature set (Model 10) produced slightly lower training AUC (0.9300) but retained stable testing AUC (0.9600) and PR-AUC (0.2900).

Interestingly, for both XGBoost and LightGBM, training on all features sometimes resulted in better testing PR-AUC than using the engineered feature set. The exact reason is unclear; it may reflect the ability of gradient boosting models to exploit subtle information in the full feature set or could be due to random variation in training. In any case, these models show strong predictive performance across feature sets and are generally more robust to feature reduction than Logistic Regression.

5. Conclusion

In this study, we proposed a structured approach to detect fraudulent transactions by leveraging RFM-based feature engineering combined with additional behavioral indicators. The RFM framework—Recency, Frequency, and Monetary value—enabled us to quantify user behavior in a way that highlights deviations from normal patterns. Recency captures how recently a customer made transactions, allowing the model to detect sudden bursts of activity that may indicate fraud. Frequency measures the transaction rate, helping to identify users who make unusually frequent transactions within short periods. Monetary features reflect transaction amounts relative to historical behavior, flagging anomalously high-value activities. By transforming raw transactional data into these interpretable behavioral features, we effectively created a lens through which the model can distinguish normal from suspicious patterns.

Beyond the RFM features, we incorporated interaction and location-based features, such as transaction-to-limit ratios, high-risk merchant categories, and cross-region transaction indicators. These additions allow the model to capture context-specific risk signals that are not apparent from RFM alone, enriching its understanding of transaction behavior.

The framework's design emphasizes both interpretability and practical utility. Instead of focusing solely on predictive metrics, we analyzed the patterns and behaviors that the model uses to identify potential fraud. This approach provides actionable insights for risk management teams, such as highlighting users with sudden spending surges or transactions outside their typical regions, even before any fraudulent activity is confirmed.

In practice, the solution achieves two key outcomes: first, it identifies anomalous transactions that warrant further investigation, reducing potential financial losses; second, it creates a scalable methodology for ongoing fraud monitoring, which can adapt to changes in user behavior over time. Overall, this study demonstrates that combining structured behavioral features with targeted contextual indicators produces a robust, interpretable, and operationally valuable fraud detection framework.

Declaration

Declaration of Pedagogical Purpose

This manuscript is prepared as part of the course project for Machine Learning and FinTech (Fall 2025) at National Yang Ming Chiao Tung University, instructed by Professor Huei-Wen Teng. The work is intended solely for pedagogical purposes, including learning, discussion, and academic training. It does not represent original research submitted for journal publication, nor should it be considered as professional financial advice.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used [ChatGPT] and [Grok], developed by [OpenAI] and[xAI], respectively, in order to improve the clarity, precision, and overall quality of the writing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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